

Unit 3. Joint, Marginal and Conditional Distributions

Two random variables: joint p.m.f./p.d.f., marginals, independence, conditioning

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 6–7, CC BY-NC-SA 4.0.

Roadmap

- 1 Why two variables?
- 2 Joint p.m.f. and p.d.f.
- 3 Marginal distributions
- 4 Independence of random variables
- 5 Conditional distributions
- 6 Wrap-up

Most economic questions are bivariate

- How does **schooling** relate to **wages**?
- How does **treatment** affect **outcomes**?
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Building blocks of this unit

Joint p.m.f./p.d.f. $\rightarrow$ Marginals $\rightarrow$ Independence $\rightarrow$ Conditional distributions.

Discrete: joint p.m.f.

Definition

For discrete (X, Y) ,

$$f_{XY}(x, y) = \mathbb{P}(X = x, Y = y), \quad \sum_{x, y} f_{XY}(x, y) = 1.$$

$$\mathbb{P}((X, Y) \in A) = \sum_{(x, y) \in A} f_{XY}(x, y).$$

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Stored as a contingency table — rows for X , columns for Y , with marginals f_X , f_Y as right and bottom margins.

Example: supermarket queues

X = customers in regular line, Y = customers in express line.

$f_{XY}(x, y)$	$y=0$	$y=1$	$y=2$	$y=3$	$y=4$	f_X
$x = 0$	0.10	0.05	0.05	0	0	0.20
$x = 1$	0.05	0.20	0.20	0.05	0	0.50
$x = 2$	0	0	0.10	0.10	0.05	0.25
$x = 3$	0	0	0	0	0.05	0.05
f_Y	0.15	0.25	0.35	0.15	0.10	1.00

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$x = 2$	0	0	0.10	0.10	0.05	0.25
$x = 3$	0	0	0	0	0.05	0.05
f_Y	0.15	0.25	0.35	0.15	0.10	1.00

Quick reads: $\mathbb{P}(X \geq 2, Y \geq 2) = 0.10 + 0.10 + 0.05 + 0 + 0 + 0.05 = 0.30$.

Continuous: joint p.d.f.

Definition

(X, Y) continuous: $f_{XY} \geq 0$ on $\mathbb{R}^2$ and

$$\mathbb{P}((X, Y) \in A) = \iint_A f_{XY}(x, y) dx dy, \quad \iint_{\mathbb{R}^2} f_{XY} = 1.$$

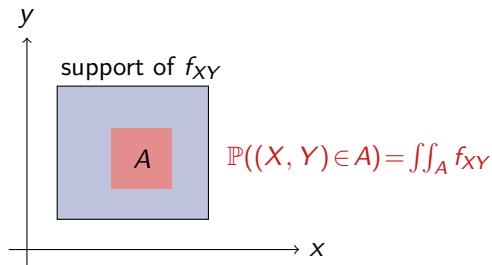
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Computing probabilities = double integration over a region.



Example: UFO over Wyoming (uniform)

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For uniform distributions, probability = area ratio. No integration needed.

Example: independent exponentials

Two spark plugs with independent exponential lifetimes:

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Both fail before time T ?

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System fails before T if both work in series? $\mathbb{P}(X + Y \leq T)$ requires integrating over the triangle $\{x + y \leq T\}$.

From joint to marginal

Marginalisation = summing/integrating out

Discrete:

$$f_X(x) = \sum_y f_{XY}(x, y), \quad f_Y(y) = \sum_x f_{XY}(x, y).$$

Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx.$$

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Marginals do not determine the joint distribution. Many joint laws share the same margins — the missing information is the **dependence structure**.

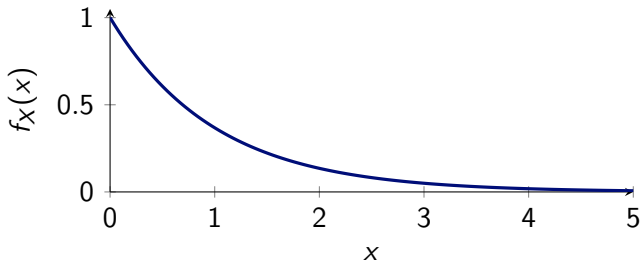
Spark plugs: marginals are exponential

$$f_{XY}(x, y) = \lambda^2 e^{-\lambda(x+y)} \Rightarrow f_X(x) = \int_0^\infty \lambda^2 e^{-\lambda(x+y)} dy = \lambda e^{-\lambda x}.$$

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Independence

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Factorisation criterion (workhorse)

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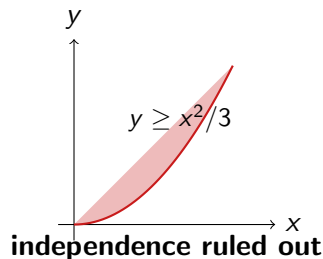
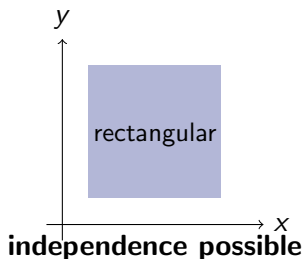
Practical shortcut. If $f_{XY}(x, y) = g(x) h(y)$ on a **rectangular support**, $X \perp Y$ even without computing the marginals.

Two diagnostic checks

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- 2 **Separable density?** On a rectangular support, independence is equivalent to $f_{XY}(x, y) = g(x)h(y)$.



Example: dependence in real data

Marriage quality $X \in \{1, 2, 3\}$, years married $Y \in \{1, 8, 12\}$. **Observed**
Independent (= product of margins)

	1	8	12	f_X
1	4.66	11.48	12.98	29.12
2	5.16	14.81	12.31	32.28
3	13.48	16.47	8.65	38.60
f_Y	23.30	42.76	33.94	100

	1	8	12	f_X
1	6.78	12.45	9.88	29.12
2	7.52	14.81	10.96	32.28
3	8.99	16.50	13.10	38.60
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The diagonal cells in the

observed table are larger than under independence; the off-diagonal cells smaller. **Dependence visible to the eye**
— testing it formally is the job of Unit 10.

Conditional p.m.f./p.d.f.

Definition

For $f_X(x) > 0$:

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- It tells us *how Y behaves once we know $X = x$* .

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Multiplication rule (already familiar): $f_{XY}(x, y) = f_{Y|X}(y | x) f_X(x)$.

Example: conditional row from a table

From the supermarket table, given $X = 1$ (so $f_X(1) = 0.50$):

y	0	1	2	3	4
$f_{XY}(1, y)$	0.05	0.20	0.20	0.05	0
$f_{Y X}(y 1)$	0.10	0.40	0.40	0.10	0

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We *re-normalise* the row by dividing by its total.

The conditional distribution sums to 1, as it must.

Example: continuous conditioning

$f_{XY}(x, y) = 2$ on the triangle $0 \leq x \leq y \leq 1$, zero elsewhere.

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Conditional distribution of Y given $X = x$:

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$Y | X = x$ is uniform on $(x, 1)$. Since this conditional depends on x , X and Y are **not independent** (consistent with non-rectangular support).

Independence revisited

Three equivalent views

The following are equivalent:

- 1 $X \perp Y$;
- 2 $f_{Y|X}(y | x) = f_Y(y)$ for every x with $f_X(x) > 0$;
- 3 $f_{X|Y}(x | y) = f_X(x)$ for every y with $f_Y(y) > 0$.

Independence revisited

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- 3 $f_{X|Y}(x | y) = f_X(x)$ for every y with $f_Y(y) > 0$.

In words: **independence means knowing X doesn't change the distribution of Y .**

This is the foundation of every “ceteris paribus” statement in econometrics.

Take-aways

- 1 Joint p.m.f./p.d.f. describes (X, Y) on $\mathbb{R}^2$; probabilities are sums or double integrals over regions.
- 2 Marginals are obtained by **summing/integrating out** the other variable. They do *not* determine the joint.
- 3 $X \perp Y$ iff $f_{XY}(x, y) = f_X(x)f_Y(y)$ on a **rectangular support**.
- 4 Conditional distribution: $f_{Y|X}(y | x) = f_{XY}(x, y)/f_X(x)$ — the foundation of regression.
- 5 Independence $\Leftrightarrow f_{Y|X}(\cdot | x)$ is the same for every x .

Next unit

Transformations of random variables: how does the distribution change when we apply a function $g(\cdot)$?

Thank you

Questions?

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